

Dynamic Programming Notes

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PROBLEM OVERVIEW

256: Exactly 3 colors.

265: There are k colors.

DP STATE

$dp[i][j]$

means

Minimum cost to paint houses 0...i such that house i is painted with color j.

TRANSITION LOGIC

Suppose current color is j.

Then previous house can be any color except j.

So

$$dp[i][j] = cost[i][j] + \min(dp[i-1][x])$$

where

$$x \neq j$$

C++ IMPLEMENTATION

```
class Solution {
public:
    int minCostII(vector<vector<int>>& costs) {

        int n = costs.size();
        if(n == 0)
            return 0;

        int k = costs[0].size();

        // dp[i][j]
        // Minimum cost to paint houses from 0...i
        // such that house i is painted with color j.
        vector<vector<int>> dp(n, vector<int>(k, 0));

        // Base Case
        //pehla ghar kaise bhi color kro
        for(int j = 0; j < k; j++) {
            dp[0][j] = costs[0][j];
        }

        // Paint remaining houses.
        for(int i = 1; i < n; i++) {

            // Try every possible color for current house.
            for(int j = 0; j < k; j++) {

                int mini = INT_MAX;

                // Previous house can have any color
                // except the current color.
                for(int prev = 0; prev < k; prev++) {

                    if(prev == j)
                        continue;

                    mini = min(mini, dp[i-1][prev]);
                }

                dp[i][j] = costs[i][j] + mini;
            }
        }

        int ans = INT_MAX;

        // Last house can end with any color
        //dp has already ensured constraints
        for(int j = 0; j < k; j++) {
            ans = min(ans, dp[n-1][j]);
        }

        return ans;
    }
};
```

```
};  
}
```